

Mathematics - Probability and Statistics

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Syllabus

- **Session 1:** Why Study Statistics?
- **Session 2:** Graphical Representation
- **Session 3:** Descriptive Statistics
- **Session 4:** Boxplots & Normal Distribution
- **Session 5:** Basic Probability & Counting Rules
- **Session 6:** Conditional Probability & Independence
- **Session 7:** Total Probability & Bayes' Theorem
- **Session 8:** Discrete Random Variables
- **Session 9:** Continuous Random Variables
- **Session 10:** Joint Distributions & Correlation
- **Session 11: Exercices/Summary**
- **Session 12:** Central Limit Theorem & Simulation
- **Session 13:** Point Estimation & MLE
- **Session 14:** Confidence Intervals
- **Session 15:** Hypothesis Testing – Basics

Total Probability & Bayes' Theorem

Session 7

Total Probability & Bayes' Theorem

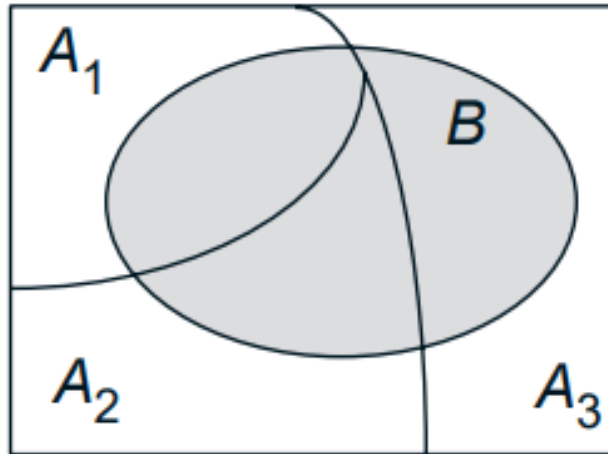
Session 7

Objectives

- Apply the Law of Total Probability to compute probabilities across partitions.
- Use probability trees to visualize conditional probabilities.
- Solve problems using Bayes' Theorem.
- Interpret results in diagnostic testing (medical tests, spam filters, reliability, etc.).

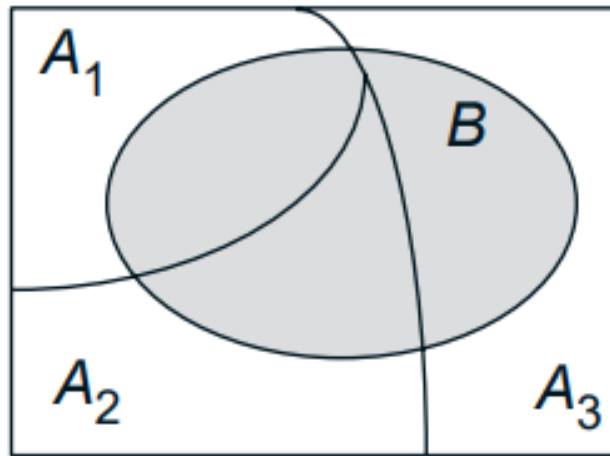
Law of Total Probability

The events $A_1, \dots, A_n$ form a partition of the sample space. How can you obtain the probability of B ?



Law of Total Probability

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B can be decomposed into the disjoint union of its intersections $A_i \cap B$ with the sets A_i :

$$B = (A_1 \cap B) \cup \dots \cup (A_n \cap B).$$

Law of Total Probability

$$B = (A_1 \cap B) \cup \dots \cup (A_n \cap B).$$

Therefore, the proba is defined by:

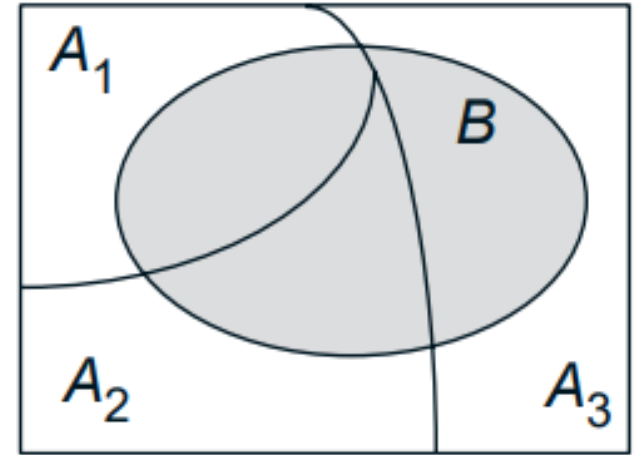
$$\mathbf{P}(B) = \mathbf{P}(A_1 \cap B) + \dots + \mathbf{P}(A_n \cap B).$$

Because we have:

$$\mathbf{P}(A_i \cap B) = \mathbf{P}(A_i)\mathbf{P}(B \mid A_i),$$

Finally, we obtain:

$$\mathbf{P}(B) = \mathbf{P}(A_1)\mathbf{P}(B \mid A_1) + \dots + \mathbf{P}(A_n)\mathbf{P}(B \mid A_n).$$



Law of Total Probability

Total Probability Theorem

Let $A_1, \dots, A_n$ be disjoint events that form a partition of the sample space (each possible outcome is included in one and only one of the events $A_1, \dots, A_n$) and assume that $\mathbf{P}(A_i) > 0$, for all $i = 1, \dots, n$. Then, for any event B , we have

$$\begin{aligned}\mathbf{P}(B) &= \mathbf{P}(A_1 \cap B) + \dots + \mathbf{P}(A_n \cap B) \\ &= \mathbf{P}(A_1)\mathbf{P}(B \mid A_1) + \dots + \mathbf{P}(A_n)\mathbf{P}(B \mid A_n).\end{aligned}$$

Law of Total Probability

Exercises:

Three machines produce **metal rods**:

- Machine 1 produces **500 rods**.
- Machine 2 produces **800 rods**.
- Machine 3 produces **700 rods**.

The machines are imperfect and produce defective rods at different rates:

- Machine 1: **2% defective**.
- Machine 2: **4% defective**.
- Machine 3: **5% defective**.

Proba of taking a defective rod **at random among all 2000 rods**.

Law of Total Probability

Sol:

- Machine 1 produces **500 rods**. $\rightarrow P(E_1)=500/2000=0.25$
- Machine 2 produces **800 rods**. $\rightarrow P(E_2)=800/2000=0.40$
- Machine 3 produces **700 rods**. $\rightarrow P(E_3)=700/2000=0.35$

If B is a event of selection a defective rod

$$P(B|E_1)= 0.02$$

$$P(B|E_2)= 0.04$$

$$P(B|E_3)= 0.035$$

$$P(B) = \sum_{i=1}^k P(B | E_i) P(E_i).$$

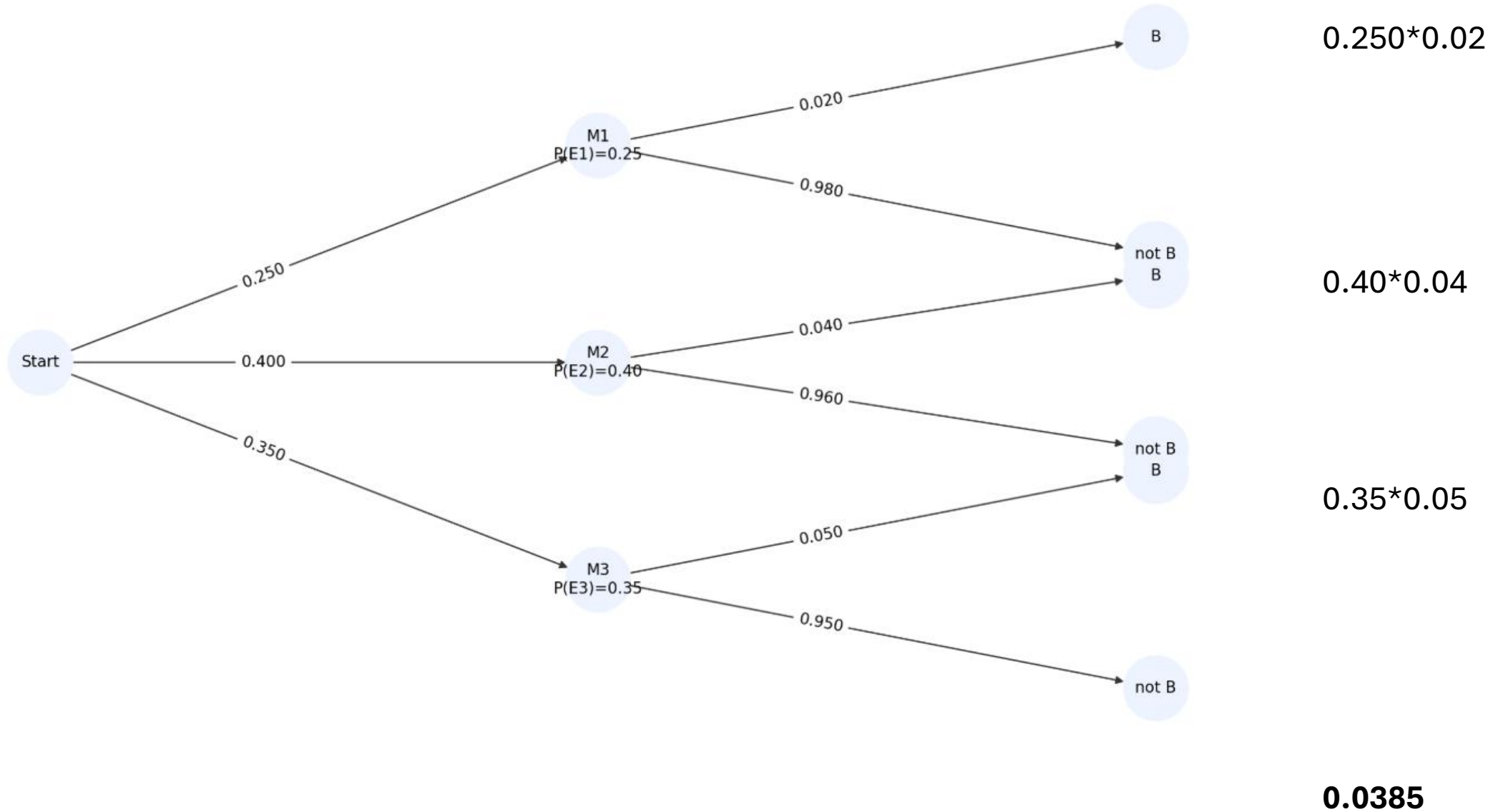
$$\begin{aligned} P(B) &= P(B|E_1)*P(E_1)+ P(B|E_2)*P(E_2)+ P(B|E_3)*P(E_3) \\ &= 0.0385 \end{aligned}$$

Tree decision

Steps

1. Root branches: $B_1, B_2, \dots, B_k$ with $P(B_i)$.
2. From each B_i , branch to A and Ac with $P(A|B_i)$ and $P(Ac|B_i)$
3. **Path probability** = product of branch probabilities.
4. $P(A)$ = **sum of path products** ending at A

Tree decision



Law of Total Probability

Exercices:

You enter a chess tournament where your probability of winning a game

- 0.3 against half the players (red players)
- 0.4 against a quarter of the players (player blue),
- 0.5 against the remaining quarter of the players (yellow players).

You play a game against a randomly chosen opponent.

What is the probability of winning?

Law of Total Probability

Sol:

You enter a chess tournament where your probability of winning a game

- 0.3 against half the players (A1)
- 0.4 against a quarter of the players (A2),
- 0.5 against the remaining quarter of the players (A3).

You play a game against a randomly chosen opponent.

What is the probability of winning?

If B is a event of winning

$$P(B|A1)= 0.3$$

$$P(B|A2)= 0.4$$

$$P(B|A3)= 0.5$$

$$P(A1)=0.5$$

$$P(A2)=P(A3)=0.25$$

$$\begin{aligned} P(B) &= P(B|A1)*P(A1)+ P(B|A2)*P(A2)+ P(B|A3)*P(A3) \\ &= 0.375 \end{aligned}$$

$$P(B) = \sum_{i=1}^k P(B | E_i) P(E_i).$$

Bayes Theorem

Problem:

A desk lamp was found to be defective. There are three factories:

Factory	% of total production	Probability of defective lamps
A	$0.35 = P(A)$	$0.015 = P(D A)$
B	$0.35 = P(B)$	$0.010 = P(D B)$
C	$0.30 = P(C)$	$0.020 = P(D C)$

If a randomly selected lamp is defective, what is the probability that the lamp was manufactured in factory C ?

Bayes Theorem

Solution:

If a randomly selected lamp is defective (D), what is the probability that the lamp was manufactured in factory C ?

$$P(C|D) = P(C \cap D) / P(D)$$

Using multiplication rule:

$$P(C|D) = P(D|C) * P(C) / P(D)$$

$$\text{As } P(D) = P(A \cap D) + P(B \cap D) + P(C \cap D)$$

And $P(A \cap D) = P(A) * P(D|A)$ and the same for B and D we have finally:

$$P(C|D) = [P(D|C) * P(C)] / [P(A) * P(D|A) + P(B) * P(D|B) + P(C) * P(D|C)]$$

$$= [0.02 * 0.3] / [(0.35 * 0.015 + 0.35 * 0.01 + 0.3 * 0.02)] = 0.407$$

Factory	% of total production	Probability of defective lamps
<i>A</i>	$0.35 = P(A)$	$0.015 = P(D A)$
<i>B</i>	$0.35 = P(B)$	$0.010 = P(D B)$
<i>C</i>	$0.30 = P(C)$	$0.020 = P(D C)$

Bayes Theorem

Bayes' theorem is very useful to solve $P(B|A)$ in terms of $P(A|B)$.

Bayes' Rule

- For any two events A and B , where $P(A) \neq 0$, we have

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}.$$

- If $B_1, B_2, B_3, \dots$ form a partition of the sample space S , and A is any event with $P(A) \neq 0$, we have

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_i P(A|B_i)P(B_i)}.$$

The probabilities $P(B)$ are often referred to as prior probabilities, because they are the probabilities of events, and that we know prior to obtaining any additional information. The conditional probabilities $P(B|A)$, and are often referred to as posterior probabilities, because they are the probabilities of the events after we have obtained additional information.

Bayes Theorem

Example:

A certain disease affects about 1 out of 10,000 people. There is a test to check whether the person has the disease. The test is quite accurate. In particular, we know that the probability that the test result is positive, given that the person does not have the disease, is only 2 percent; the probability that the test result is negative, given that the person has the disease, is only 1 percent.

A random person gets tested for the disease and the result comes back positive. What is the probability that the person has the disease?

Bayes Theorem

Solution:

D=event that the person has the disease
T= event that the test result is true/positive.

$$P(D)=1/10000$$

$$P(T|D^c)=0.02$$

$$P(T^c|D)=0.01 \rightarrow P(T|D)=0.99$$

We want $P(D|T)$?

Bayes' Theorem:

$$P(D|T)=P(T|D)*P(D)/P(T)$$

$$P(T)=P(T|D)*P(D)+P(T|D^c)*P(D^c)=0.99*1/10000+0.02*9999/10000= 0.020097$$

$$\text{So: } P(D|T)=0.99*1/10000/ 0.020097= 0.0049=\mathbf{0.5\%}$$

$$P(B) = \sum_{i=1}^k P(B | E_i) P(E_i).$$

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}.$$

Exercise

Exercise:

Alice is taking a probability class and at the end of each week she can be either up-to-date or she may have fallen behind. If she is up-to-date in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.8 (or 0.2, respectively). If she is behind in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.6 (or 0.4, respectively). Alice is up-to-date when she starts the class. What is the probability that she is up-to-date after three weeks?

Exercise

Solution:

Alice is taking a probability class and at the end of each week she can be either up-to-date or she may have fallen behind. If she is up-to-date in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.8 (or 0.2, respectively). If she is behind in a given week, the probability that she will be up-to-date (or behind) in the next week is 0.6 (or 0.4, respectively). Alice is up-to-date when she starts the class. What is the probability that she is up-to-date after three weeks?

Let U_i and B_i be the events that Alice is **up-to-date or behind**, respectively, after i weeks. According to the total probability theorem, the desired probability $P(U_3)$ is given by

- $P(U_3) = P(U_2)P(U_3 | U_2) + P(B_2)P(U_3 | B_2) = P(U_2) \cdot 0.8 + P(B_2) \cdot 0.6$
- $P(U_2) = P(U_1)P(U_2 | U_1) + P(B_1)P(U_2 | B_1) = P(U_1) \cdot 0.8 + P(B_1) \cdot 0.6$
- $P(B_2) = P(U_1)P(B_2 | U_1) + P(B_1)P(B_2 | B_1) = P(U_1) \cdot 0.2 + P(B_1) \cdot 0.4.$

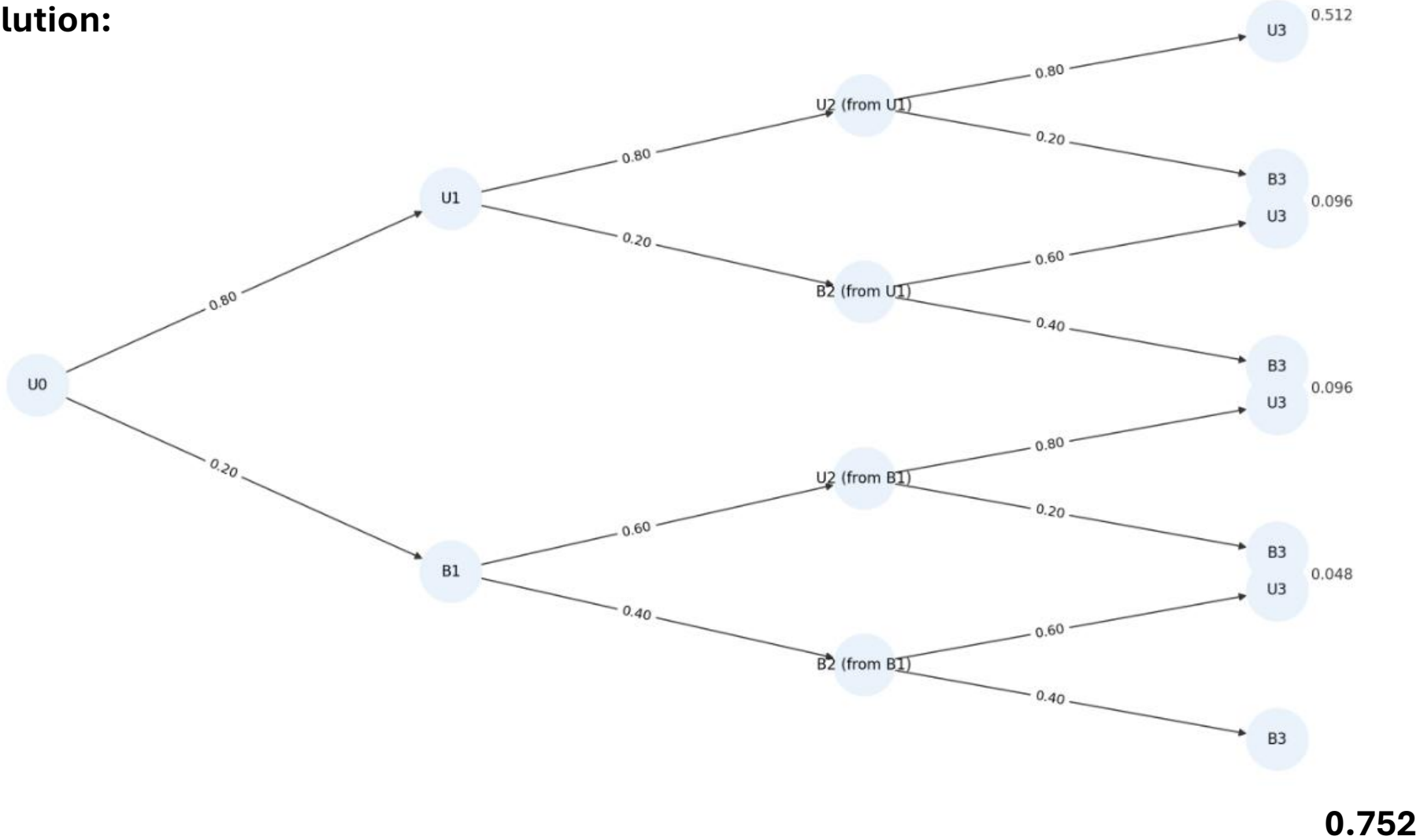
$$P(B) = \sum_{i=1}^k P(B | E_i) P(E_i).$$

$$P(U_2) = 0.8 \cdot 0.8 + 0.2 \cdot 0.6 = 0.76$$

$$P(B_2) = 0.8 \cdot 0.2 + 0.2 \cdot 0.4 = 0.24 \quad \text{therefore: } P(U_3) = 0.76 \cdot 0.8 + 0.24 \cdot 0.6 = \mathbf{0.752}$$

Exercise

Solution:



Exercise

Problem:

An insurance company believes that people can be divided into two classes — those that are accident prone and those that are not. Their statistics show that an accident-prone person will have an accident at some time within a fixed 1-year period with probability .4, whereas this probability decreases to .2 for a non-accident-prone person.

If we assume that 30 percent of the population is accident prone, what is the probability that a new policy holder will have an accident within a year of purchasing a policy?

Exercise

Solution: If we assume that 30 percent of the population is accident prone, what is the probability that a new policy holder will have an accident within a year of purchasing a policy?

D=Drivers prone to have accident

$P(D)=0.3$

A=having accident with a year

$P(A|D)=0.4$

$P(A|D^c)=0.2$

$P(A)=P(A|D^c)*P(D^c)+P(A|D)*P(D)=$
 $=0.2*(1-0.3)+0.4*0.3$
 $=\mathbf{0.26}$

$$P(B) = \sum_{i=1}^k P(B | E_i) P(E_i).$$

Exercise

Problem:

The manual of a product states that the lifetime T , is defined as the amount of time (in years) the product works properly until it breaks down, and satisfies

$$P(T \geq t) = e^{-t/5}, \text{ for all } t \geq 0.$$

I purchase the product and use it for two years without any problems. What is the probability that it breaks down in the third year?

Exercise

Solution:

$$P(T \geq t) = e^{-t/5}, \text{ for all } t \geq 0.$$

I purchase the product and use it for two years without any problems. What is the probability that it breaks down in the third year

A= event that a purchased product breaks down in the third year

B= event that a purchased product does not break down in the first two years

We want: $P(A|B)$

$$P(A|B) = P(B|A)P(A)/P(B)$$

$$P(B) = e^{-2/5} = 0.6703$$

$$P(A) = P(2 < T \leq 3) = e^{-2/5} - e^{-3/5} = 0.6703 - 0.548 = 0.122$$

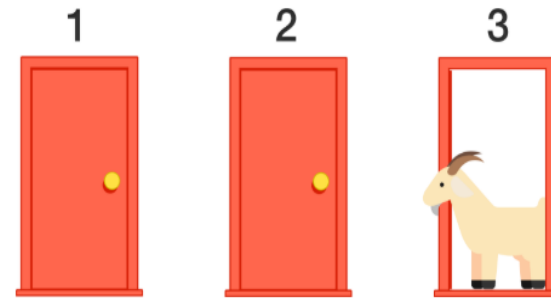
$P(B|A)$ = proba that the event does break during the first 2 years if it breaks the third year.....=1

$$P(A|B) = 1 * 0.122 / 0.6703 = 0.182 = \mathbf{18\%}$$

Exercise

Problem: Monty Hall Problem

In the problem, you are on a game show, being asked to choose between **three doors**. Behind each door, there is either a car or a goat (2 goat 1 car). **You choose a door. The host, picks one of the other doors, which he knows has a goat behind it, and opens it, showing you the goat.** The host then asks whether you would like to switch your choice of door to the other remaining door.



Assuming you prefer having a car more than having a goat, do you choose to switch or not to switch?

What is the odd if you switch? What is the odd if you do not switch?

Exercise

Solution:

From the decision tree, we see that the probability
To have a car is $\frac{2}{3}$ if you switch while only $\frac{1}{3}$
If you stick.

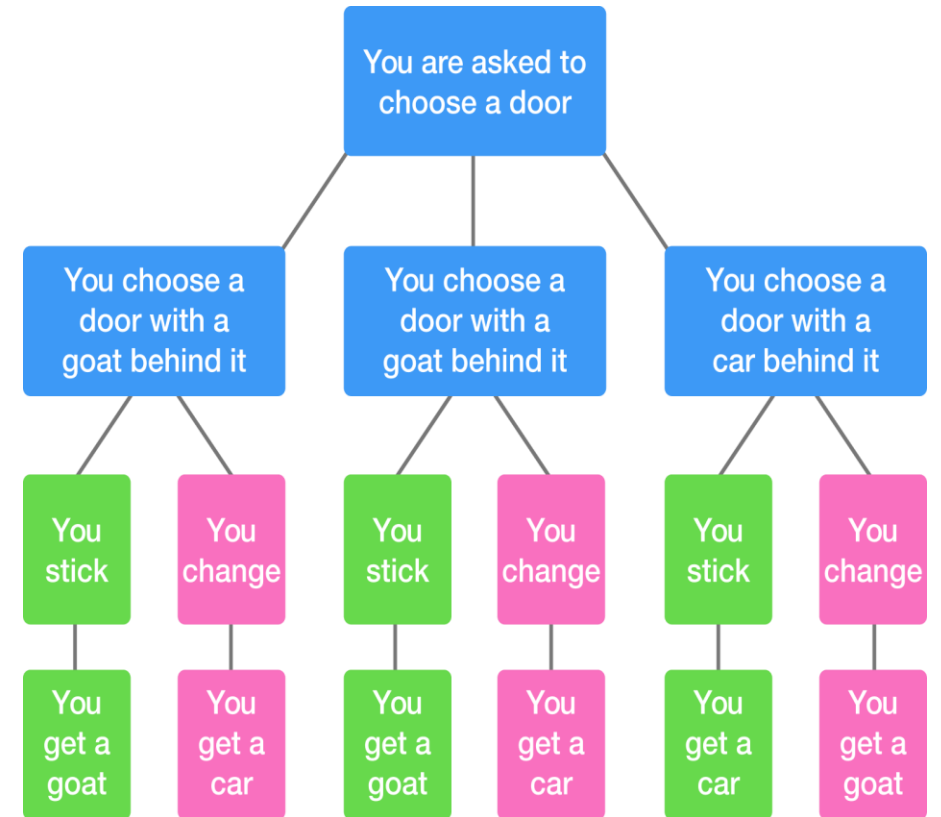
Now let use Bayes theorem. With this theorem we can find if the
probability that our initially picked door is the one with the car behind it
given that the host has opened a door showing a goat behind it

Imagine you select door 1:

H =hypothesis "door 1 has a car behind it

E = evidence that host has revealed a door with a goat behind it

**We want to find $P(H|E)$ and will compare with the probability
Is we change the door.**



Exercise

Solution:

$$P(H | E) = \frac{P(E | H)}{P(E)} P(H) = \frac{P(E | H) \times P(H)}{P(E | H) \times P(H) + P(E | \text{not}H) \times P(\text{not}H)}.$$

- $P(H)$ is the prior probability that door 1 has a car behind it, without knowing about the door that Monty reveals. This is $\frac{1}{3}$.
- $P(\text{not}H)$ is the probability that we did not pick the door with the car behind it. Since the door either has the car behind it or not, $P(\text{not}H) = 1 - P(H) = \frac{2}{3}$.
- $P(E | H)$ is the probability that Monty shows a door with a goat behind it, given that there is a car behind door 1. Since Monty always shows a door with a goat, this is equal to 1.
- $P(E | \text{not}H)$ is the probability that Monty shows the goat, given that there is a goat behind door 1. Again, since Monty always shows a door with a goat, this is equal to 1.

$$P(H | E) = \frac{1 \times \frac{1}{3}}{1 \times \frac{1}{3} + 1 \times \frac{2}{3}} = \frac{\frac{1}{3}}{1} = \frac{1}{3}.$$

The probability that the car is behind door 1 is completely unchanged by the evidence. However, since the car can only either be behind door 1 or behind the door Monty didn't reveal, the probability it is behind the door that is not revealed is $1 - 1/3 = \mathbf{2/3}$. Therefore, switching is twice as likely to get you the car as staying!!

Exercise

Problem:

. Each bag in a large box contains 25 tulip bulbs. Three-fourths of the bags are of Type A containing bulbs for 5 red and 20 yellow tulips; one-fourth of the bags are of Type B contain bulbs for 15 red and 10 yellow tulips. If the tulip is red, what is the probability that bag B was selected?

Exercise

Solution:

Each bag in a large box contains 25 tulip bulbs. Three-fourths of the bags are of Type A containing bulbs for 5 red and 20 yellow tulips; one-fourth of the bags are of Type B contain bulbs for 15 red and 10 yellow tulips. If the tulip is red, what is the probability that bag B was selected?

A=Type A. $P(A)=3/4$

B= Type B $P(B)=1/4$

R= tulip red

We want $P(B|R) = P(R|B)P(B)/P(R)$

- $P(R|B)=15/25=3/5$ and $P(R|A)=1/5$

$P(R) = P[(A \cap R) \cup (B \cap R)]$

Using multiplicative rule and mutually exclusive events

$$P(B) = \sum_i P(B | E_i)P(E_i).$$

$P(R)=P(R|A)P(A)+P(R|B)P(B)=1/5*0.75+0.25*3/5=0.30$

$P(B|R)=3/5*1/4/(0.30)=\mathbf{0.5}$

Hypothesis Testing – Basics

Session 15

Hypothesis Testing – Basics

Session 15

Objectives:

- State and formalize null (H_0) and alternative (H_1) hypotheses; choose one- vs two-sided.
- Explain Type I / Type II errors, significance level α , power ($1-\beta$), p-value.
- Execute hypothesis tests

Hypothesis Testing : What is it?

In the previous chapter, we demonstrated how to build a confidence interval estimate of a parameter using sample data. However, in many engineering problems, the task is not just to estimate a parameter but to determine which of two competing statements about it is true. These statements are called ***hypotheses***, and the process of making such decisions is known ***as hypothesis testing***.

Examples: drug efficacy, manufacturing tolerances, A/B tests

Hypothesis Testing : Theory

Every time we perform a hypothesis test, this is the basic procedure that we will follow:

- State the research hypothesis and the null hypothesis

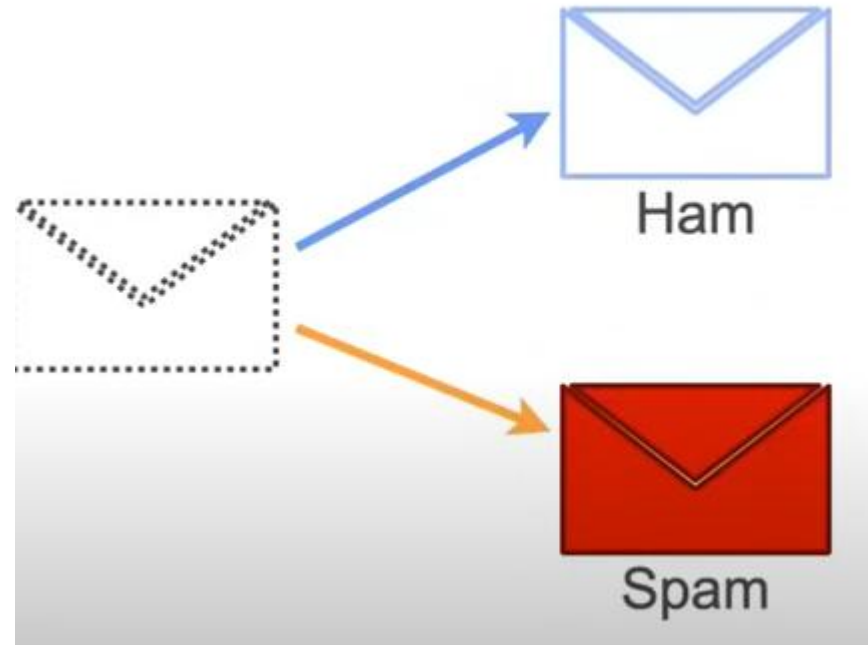
Null hypothesis (H_0) is a statistical assumption suggesting that any observed patterns or differences in a dataset are due to chance and not the result of a meaningful effect.

The other hypothesis is called the alternative hypothesis (H_1). Both are mutually exclusive.

- The null hypothesis is generally an equality and the alternative hypothesis might be either one-sided or two-sided, depending on the conclusion to be drawn if H_0 is rejected.
- We'll make an initial assumption about the population parameter.
- We'll collect evidence or else use somebody else's evidence (data).
- Based on the available evidence (data), we'll decide whether to "**reject**" or "**not reject**" our initial assumption.

Hypothesis Testing : example

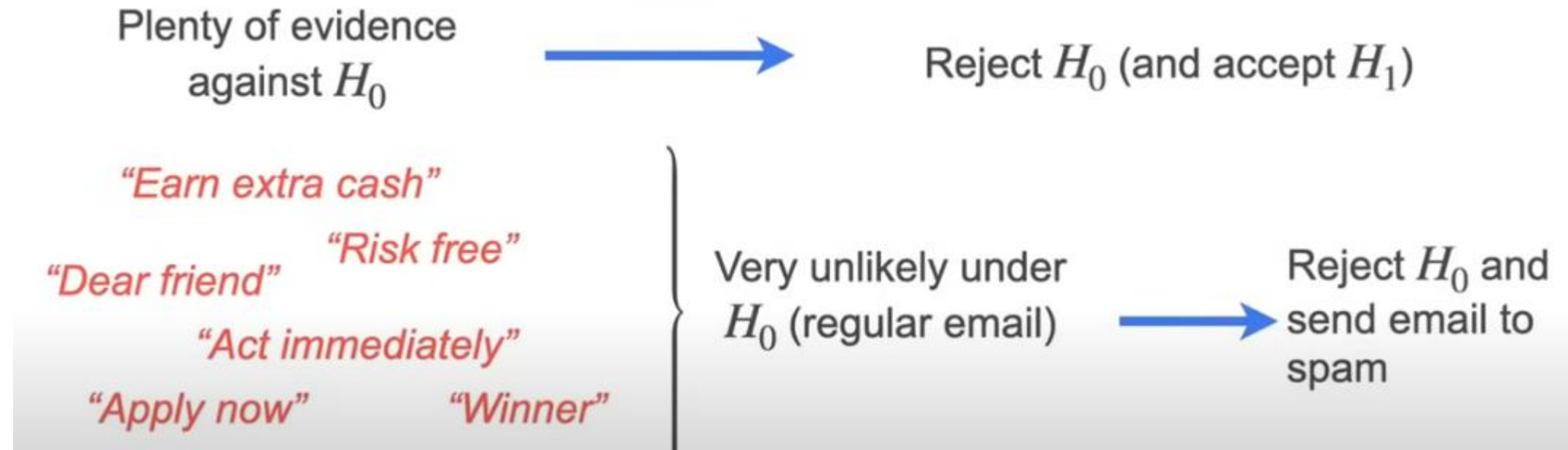
Example: You have a spam email detector, and its role is to say if the email is ham or spam



Hypothesis Testing : example

Example: By default you assumed that all emails are ham because it is worst to delete a good email than to have a spam!!

Therefore, our base assumption is that an email is ham (H_0), meaning that nothing is happening.



Hypothesis Testing : errors

Decision	Reality	
	H_0 True (Not spam)	H_0 False (Spam)
Reject H_0 (Decide spam)	Type I error	Correct
Don't reject H_0 (Decide not spam)	Correct	Type II error

Rejecting the null hypothesis H_0 when it is true is defined as a **type I error**.

Failing to reject the null hypothesis when it is false is defined as a **type II error**.

significance level, or the α -error: $\alpha = P(\text{type I error}) = P(\text{reject } H_0 \text{ when } H_0 \text{ is true})$

the β -error: $\beta = P(\text{type II error}) = P(\text{fail to reject } H_0 \text{ when } H_0 \text{ is false})$

Hypothesis Testing : Making a decision

There are two approaches to making the decision:

- one is called the "**critical value**" approach
- and the other is called the "**p-value**" approach

Hypothesis Testing : critical value

The basic procedure is:

1. State the **null hypothesis** H_0 and the **alternative hypothesis** H_1

2. collect evidence (data)

3. Calculate the **test statistic**:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

4. Determine the **critical region**.

5. **Make a decision**. Determine if the test statistic falls in the critical region. If it does, reject the null hypothesis. If it does not, do not reject the null hypothesis.

Now, back to those possible errors we can make when conducting such a hypothesis test.

Hypothesis Testing : critical value

Example: A four-sided die is tossed 1000 times, and 290 fours are observed. Is there evidence to conclude that the die is biased, that is, say, that more fours than expected are observed?

1.) **H₀:** Die is not biased, therefore the probability to have a 4 is 0.25

H₀: p=0.25

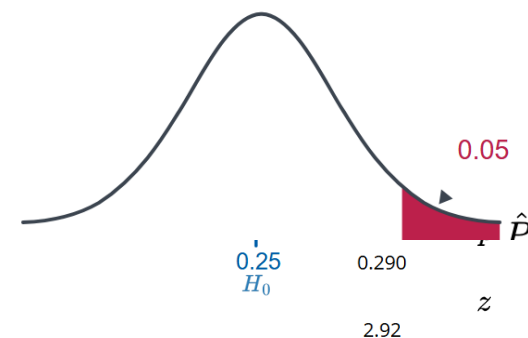
2.) The die was tossed 1000 times, and 290 fours were observed. So from this sample we have $\hat{p}=290/1000=0.29$

3.)
$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$
 with $n=1000$ you find $Z=2.92$

4.) **Critical value approach** tells us to define a threshold value, called a "**critical value**"
 $\alpha=0.05$ therefore, $z_{\text{crit}} = z_{0.95} = 1.645$. If $Z > z_{\text{crit}}$, we reject H₀

5.) **Make a decision:** $Z=2.92 > z_{\text{crit}}=1.645$

we reject the null hypothesis at the 5% significance level and conclude with at least 95% confidence that the die is biased



Hypothesis Testing : critical value

Example: Let p equal the proportion of drivers who use a seat belt in a country that does not have a mandatory seat belt law. It was claimed that $p=0.14$. An advertising campaign was conducted to increase this proportion. Two months after the campaign, $y=104$ out of a random sample of $n=590$ drivers were wearing seat belts.

Was the campaign successful? (use a significance level of 0.01)

Hypothesis Testing : critical value

Solution:

1.) $H_0: p_0=0.14$

2.) $y=104$ over $n=590$. From this sample we have $\hat{p}=104/590=0.176$

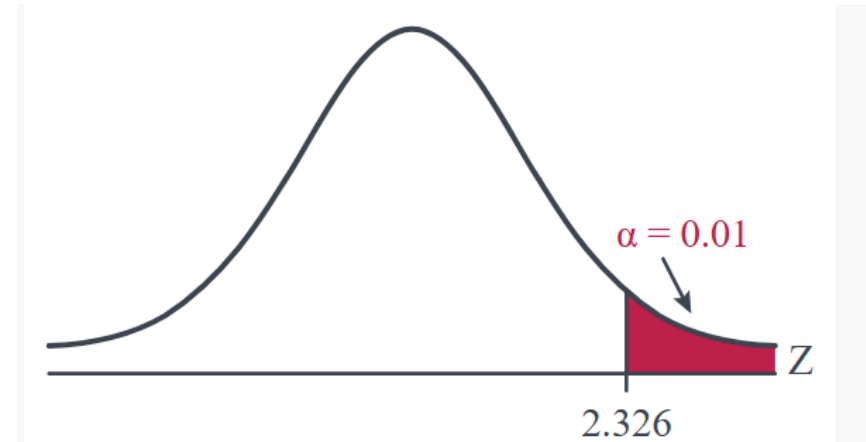
$H_1: p > P_0=0.14$

3.) $Z=2.52$

4.) With $\alpha=0.01$, you have $z_{\text{crit}} = z_{0.99}=2.326$

5.) $Z > z_{\text{crit}}$,

we reject the null hypothesis at the 1% significance level and conclude with at least 99% confidence that the campaign was successful ($p > 0.14$)



Hypothesis Testing : critical value

Example: 47% of the 1052 U.S. adults surveyed classified themselves as "very happy" when given the choices of:

"very happy"

"fairly happy"

"not too happy"

Suppose that a journalist who is a pessimist took advantage of this poll to write a headline titled "Poll finds that U.S. adults who are very happy are in the minority." Is the pessimistic journalist's headline warranted? (use a significance level of 0.05)

Hypothesis Testing : critical value

Solution:

1.) $H_0: p_0=0.5$

2.) $\hat{p}=0.47$

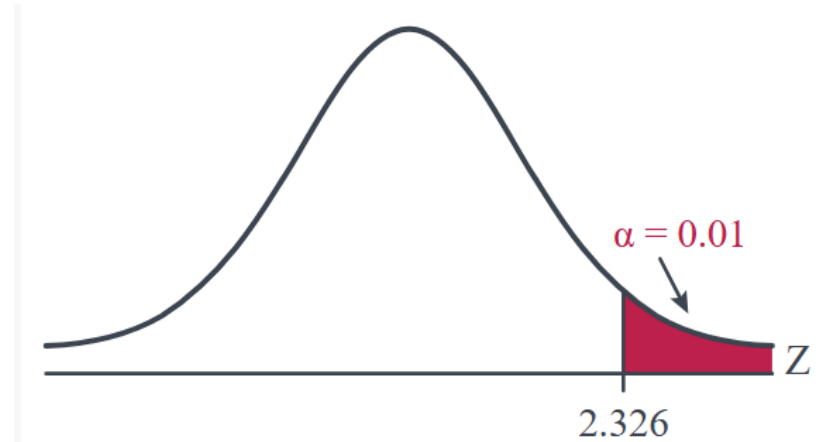
3.) $Z=-1.946$

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

4.) With $\alpha=0.05$, you have $z_{\text{crit}} = z_{0.05} = -1.645$

5.) $Z < z_{\text{crit}}$,

we cannot reject the null hypothesis at the 5% significance level and conclude with at least 95% confidence that adults who are very happy are in the minority



Hypothesis Testing : p-value

For some example, we rejected the null hypothesis when $\alpha=0.05$ but, we failed to reject the null hypothesis when $\alpha=0.01$. There must be some level of α , then, in which we cross the threshold from rejecting to not rejecting the null hypothesis. What is the smallest that would still cause us to reject the null hypothesis? **This is the p-value approach!!**

Definition: The Probability of Rejecting a Null Hypothesis (H_0), the hypothesis that there is no relationship between two measured phenomena

- In other word, the p-value measures how confidence we are about the assumption given the observed data. It ranges from 0 to 1. The smaller the p -value, the less likely the results occurred by random chance, and the stronger the evidence that you should reject the null hypothesis
- A p-value of 0.05 is commonly referred to as the level of significance (α).
If the p-value > 0.05 : It indicates that the null hypothesis should be accepted.
If the p-value < 0.05 : It indicates that the null hypothesis should be rejected.

A p-value of 0.05 means there is only a 5 % chance of obtaining a result at least as extreme as the one observed, assuming the null hypothesis is correct.

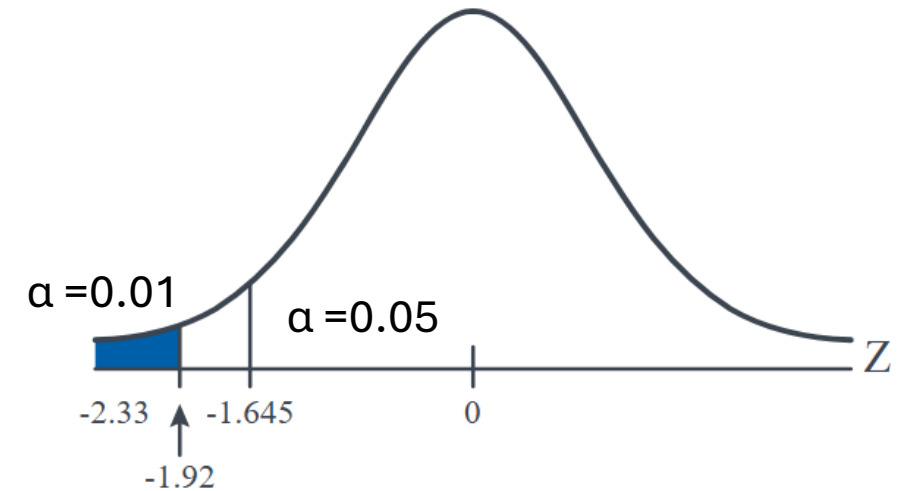
Hypothesis Testing : p-value

Example: Among patients with lung cancer, usually, 90% or more die within three years. As a result of new forms of treatment, it is felt that this rate has been reduced. In a recent study of $n = 150$ lung cancer patients, $y = 128$ died within three years.

- $H_0: p=0.90$, $H_1: p<0.90$
- Statistic test:

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.853 - 0.90}{\sqrt{\frac{0.90(0.10)}{150}}} = -1.92$$

- Using Z table, we found a **p-value of 0.0274**



There is only a **2.74%** chance of seeing a sample this extreme (or more) if the true death rate were **90%**, therefore

Hypothesis Testing : Equality of Two Means

When Population Variances Are Equal (pooled t-test)

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{S_p \sqrt{\frac{1}{n} + \frac{1}{m}}}$$

follows a t_{n+m-2} distribution where S_p^2 , the pooled sample variance:

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$$

When Population Variances Are Not Equal (Welch's t-test)

Let's again start with the good news that we've already done the dirty theoretical work here. Recall that if you have two independent samples from two normal distributions with unequal variances $\sigma_X^2 \neq \sigma_Y^2$, then:

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}}$$

follows, at least approximately, a t_r distribution where r , the adjusted degrees of freedom is determined by the equation:

$$r = \frac{\left(\frac{s_X^2}{n} + \frac{s_Y^2}{m} \right)^2}{\frac{(s_X^2/n)^2}{n-1} + \frac{(s_Y^2/m)^2}{m-1}}$$

If r doesn't equal an integer, as it usually doesn't, then we take the integer portion of r . That is, we use $\lfloor r \rfloor$ if necessary.

Hypothesis Testing : Equality of Two Means

Example: Is the mean fastest speed driven by male college students different than the mean fastest speed driven by female college students?

Males (X)	Females (Y)
$n = 34$	$m = 29$
$\bar{x} = 105.5$	$\bar{y} = 90.9$
$s_x = 20.1$	$s_y = 12.2$

Hypothesis Testing : Equality of Two Means

Solution: Is the mean fastest speed driven by male college students different than the mean fastest speed driven by female college students?

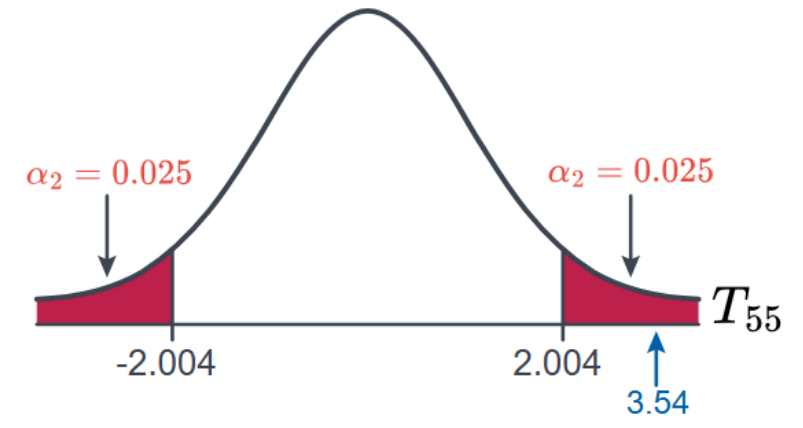
Two-sided:

$$H_0 : \mu_M - \mu_F = 0$$

$$t = \frac{(105.5 - 90.9) - 0}{\sqrt{\frac{20.1^2}{34} + \frac{12.2^2}{29}}} = 3.54$$

$$r = \frac{\left(\frac{12.2^2}{29} + \frac{20.1^2}{34}\right)^2}{\left(\frac{1}{28}\right)\left(\frac{12.2^2}{29}\right)^2 + \left(\frac{1}{33}\right)\left(\frac{20.1^2}{34}\right)^2} = 55.5$$

Males (X)	Females (Y)
$n = 34$	$m = 29$
$\bar{x} = 105.5$	$\bar{y} = 90.9$
$s_x = 20.1$	$s_y = 12.2$



Then, the **critical value approach** tells us to reject the null hypothesis in favor of the alternative hypothesis if:

$$t > t_{0.025, 55} = 2.004$$

And again, the decision is the same using the p -value approach. The p -value is 0.0008:

$$P = 2 \times P(T_{55} > 3.54) = 2(0.0004) = 0.0008$$

Therefore, because $p = 0.008 \leq \alpha = 0.05$, we reject the null hypothesis in favor of the alternative hypothesis. Again, we conclude that there is sufficient evidence at the $\alpha = 0.05$ level to conclude that the average fastest speed driven by the population of male college students differs from the average fastest speed driven by the population of female college students.

Hypothesis Testing : Comparing Two Proportions

The test statistic for testing the difference in two population proportions, that is, for testing the null hypothesis $H_0 : p_1 - p_2 = 0$ is:

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

where:

$$\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2}$$

the proportion of "successes" in the two samples combined.

Hypothesis Testing : Comparing Two Proportions

Exercise: Is there sufficient evidence at $\alpha=0.05$, to conclude that the two populations — smokers and non-smokers — differ significantly with respect to their opinions?

Non- Smokers	Smokers
$n_1 = 605$	$n_2 = 195$
$y_1 = 351$ said "yes"	$y_2 = 41$ said "yes"
$\hat{p}_1 = \frac{351}{605} = 0.58$	$\hat{p}_2 = \frac{41}{195} = 0.21$

Hypothesis Testing : Comparing Two Proportions

Solution: p_1 =the proportion of the non-smoker population who reply "yes" and p_2 = the proportion of the smoker population who reply "yes," then we are interested in testing the null hypothesis:

- Two-sided: $H_0: p_1 = p_2$ and $H_1: p_1 \neq p_2$
- Statistic test

$$\hat{p} = \frac{41 + 351}{195 + 605} = \frac{392}{800} = 0.49$$

$$Z = \frac{(0.58 - 0.21) - 0}{\sqrt{0.49(0.51) \left(\frac{1}{195} + \frac{1}{605} \right)}} = 8.99$$

- Z is >>>>1.96 (alpha=0.05) so we can reject the H_0 hypothesis. There is sufficient evidence at the 0.05 level to conclude that the two populations differ with respect to their opinions.

Non- Smokers	Smokers
$n_1 = 605$	$n_2 = 195$
$y_1 = 351$ said "yes"	$y_2 = 41$ said "yes"
$\hat{p}_1 = \frac{351}{605} = 0.58$	$\hat{p}_2 = \frac{41}{195} = 0.21$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2}$$

Hypothesis Testing

Exercices: Boys of a certain age are known to have a mean (μ) weight of 85 pounds. A complaint is made that the boys living in a municipal children's home are underfed. As one bit of evidence, $n=25$ boys (of the same age) are weighed and found to have a mean weight of $\bar{x} = 80.94$ pounds. It is known that the population standard deviation is 11.6 pounds. Based on the available data, what should be concluded concerning the complaint?

Hypothesis Testing

Solutions: Boys of a certain age are known to have a mean (μ) weight of 85 pounds. A complaint is made that the boys living in a municipal children's home are underfed. As one bit of evidence, $n=25$ boys (of the same age) are weighed and found to have a mean weight of $\bar{x} = 80.94$ pounds. It is known that the population standard deviation is 11.6 pounds. Based on the available data, what should be concluded concerning the complaint?

$H_0: \mu=85$, alternative $H_1: \mu<85$

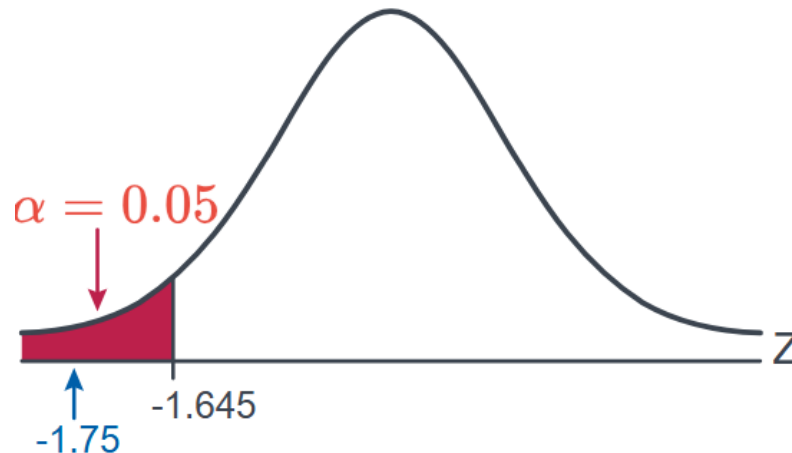
The variance population is known, therefore, we can apply Z test.

$$Z = (80.94 - 85) / [11.6 / \sqrt{25}] = -1.75$$

For $\alpha=0.05$, $z=-1.65$

$Z < z$, we can reject H_0

Using p-value=0.0401



1. $X_1, X_2, \dots, X_n$ is a random sample from a normal population with mean μ and variance σ^2 . So that:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ and } Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

2. The population variance σ^2 is known.

Then, a $(1 - \alpha)100\%$ confidence interval for the mean μ is:

$$\bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

he interval, because it depends on Z , is often referred to as the Z -interval for a mean.

Hypothesis Testing

Exercices: A manufacturer of hard safety hats for construction workers is concerned about the mean and the variation of the forces its helmets transmits to wearers when subjected to an external force. The manufacturer has designed the helmets so that the mean force transmitted by the helmets to the workers is 800 pounds (or less) with a standard deviation to be less than 40 pounds. Tests were run on a random sample of $n = 40$ helmets, and the sample mean and sample standard deviation were found to be 825 pounds and 48.5 pounds, respectively. Do the data provide sufficient evidence, at the $\alpha=0.05$ level, to conclude that the population standard deviation **exceeds** 40 pounds?

Hypothesis Testing

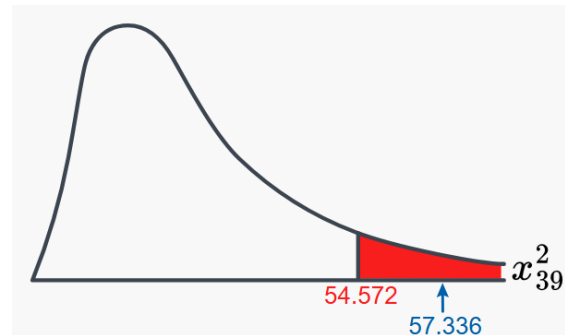
Solution. The manufacturer has designed the helmets so that the mean force transmitted by the helmets to the workers is 800 pounds (or less) with a standard deviation to be less than 40 pounds. Tests were run on a random sample of $n = 40$ helmets, and the sample mean and sample standard deviation were found to be 825 pounds and 48.5 pounds, respectively.

Do the data provide sufficient evidence, at the $\alpha=0.05$ level, to conclude that the population standard deviation **exceeds** 40 pounds?

$$H_0: \sigma^2 = 40^2 = 1600 \quad H_1: \sigma^2 > 1600$$

You have random sample of size n from a **normal population** with (unknown) mean μ and variance σ^2 .

$$\chi^2 = \frac{(40 - 1)48.5^2}{40^2} = 57.336$$



$$\chi^2 = \frac{(n - 1)S^2}{\sigma_0^2}$$

Hypothesis Testing

Exercices: Is there sufficient evidence at the $\alpha=0.05$ level to conclude that the variance of the fastest speed driven by male college students differs from the variance of the fastest speed driven by female college students?

Males (X)	Females (Y)
$n = 34$	$m = 29$
$\bar{x} = 105.5$	$\bar{y} = 90.9$
$s_x = 20.1$	$s_y = 12.2$

Hypothesis Testing

Solution: Is there sufficient evidence at the $\alpha=0.05$ level to conclude that the variance of the fastest speed driven by male college students differs from the variance of the fastest speed driven by female college students?

$$H_0: \sigma_X^2 = \sigma_Y^2 \quad \text{and} \quad H_1: \sigma_X^2 \neq \sigma_Y^2$$

we can use the test statistic:

$$F = \frac{S_X^2}{S_Y^2} \quad F = \frac{12.2^2}{20.1^2} = 0.368$$

Now, we check the value in the F table using

$$F_{1-(\alpha/2)}(n-1, m-1) = \frac{1}{F_{\alpha/2}(m-1, n-1)}$$

Because the test statistic falls in the rejection region, we reject the null hypothesis in favor of the alternative hypothesis. There is sufficient evidence at the $\alpha=0.05$ level to conclude that the population variances are not equal

Males (X)	Females (Y)
$n = 34$	$m = 29$
$\bar{x} = 105.5$	$\bar{y} = 90.9$
$s_x = 20.1$	$s_y = 12.2$

